

2024, Mar. 2024 Release LOC:Acute Pediatric Anemia

Overview

Select Day

Episode Day 1

Episode Day 2

Episode Day 3

Discharge Screens ⁽²³⁾

Notes

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Overview

Instruction:

This subset is for patients with anemia. For patients with gastrointestinal bleeding, refer to the Gastrointestinal (GI) Bleeding (Pedi) subset. For patients with epistaxis, uterine or vaginal bleeding (excluding pregnancy), hemophilia or von Willebrand disease, or hemoptysis, refer to the Non-traumatic Bleeding (Pedi) subset.

Introduction:

Anemia is defined as a reduction in hemoglobin or red blood cell (RBC) mass at least two standard deviations below the mean, based on age and sex assigned at birth (Park, In: Elsevier, ed. The Harriet Lane Handbook, 23 edn. 2024:365-407). In children, anemia can be categorized as decreased production or increased RBC turnover. RBC turnover can be due to increased hemolysis (intrinsic factors of the RBC such as medicine, glucose-6-phosphate dehydrogenase (G6PD) deficiency, sickle cell disease, mechanical destruction) or RBC losses (internal or external bleeding). Increase in RBC turnover may be caused by blood loss, mechanical destruction or hemolysis, which may be the result of inherited defects in RBCs, medications, mechanical valves, or splenomegaly. Iron deficiency is a common cause of decreased RBC production and can be treated in the outpatient setting. Additional causes may include inflammation from chronic infections or other conditions such as renal failure, medication use, viral illness, and bone marrow disorders (Park, In: Elsevier, ed. The Harriet Lane Handbook, 23 edn. 2024:365-407; Verbillion and Dupre, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:1462-79; Long and Koyfman, Emerg Med Clin North Am 2018, 36: 609-30).

Evaluation and Treatment:

The evaluation of patients with anemia includes determining the RBC morphology. Treatment for anemia is based on the cause. Patients may require blood transfusion or treatment specific to the cause of bleeding (Park, In: Elsevier, ed. The Harriet Lane Handbook, 23 edn. 2024:365-407; Long and Koyfman, Emerg Med Clin North Am 2018, 36: 609-30).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:**● Episode Day 1, One:**

(Symptom or finding within 24h)

● OBSERVATION, One: ⁽¹⁾

- Gastrointestinal (GI) bleeding (use **Gastrointestinal Bleeding subset**)
- Non-traumatic bleeding (use **Non-Traumatic Bleeding subset**)

● Anemia and, Both:**● Finding, ≥ One:**

- Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) ⁽²⁾
- Exertional dyspnea ⁽³⁾

● Orthostatic hypotension, Both:

- Sustained drop in blood pressure within 3 min of sitting or standing
- Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
- Syncope ⁽⁴⁾

● Intervention, Both:

- Blood product transfusion ⁽²⁾
- Hct or Hb monitoring at least daily

● CRITICAL, ≥ One: ⁽⁵⁾**● Hemodynamic instability, Both:****● Finding, ≥ Two:****● Heart rate, One:** ⁽⁶⁾

- > 180/min, sustained (age ≤ 12 mos) ⁽⁷⁾
- > 160/min, sustained (age > 12 mos to 4 yrs) ⁽⁷⁾
- > 150/min, sustained (age > 4 yrs to 8 yrs) ⁽⁷⁾
- > 140/min, sustained (age > 8 yrs to 15 yrs) ⁽⁷⁾
- > 130/min, sustained (age > 15 yrs to < 18 yrs) ⁽⁷⁾

● Systolic BP, One:

- < 70 mmHg (age ≤ 12 mos)
- < 80 mmHg (age > 1 to < 5 yrs)
- < 90 mmHg (age 5 to < 18 yrs)
- Labile ⁽⁸⁾

● MAP, One: ⁽⁹⁾

- < 55 mmHg (≤ 12 mos)
- < 65 mmHg (> 1 to < 18 yrs)

● Intervention, ≥ One:

- Antifibrinolytic ⁽¹⁰⁾
- Blood product transfusion ⁽¹¹⁾
- IV fluid resuscitation ≤ 2d ⁽¹²⁾
- Hemodynamic monitoring, invasive ⁽¹³⁾
- Mechanical ventilation
- NIPPV ⁽¹⁴⁾
- Oxygen ≥ 40%(0.40) ⁽¹⁵⁾
- Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● Episode Day 2, One:**● OBSERVATION, One:****● Responder, discharge expected today if clinically stable last 12h, Both:** ⁽¹⁶⁾

- Hct or Hb within acceptable limits ⁽¹⁷⁾
- Vital signs within acceptable limits ⁽¹⁷⁾

● Partial responder, not clinically stable for discharge and requires continued stay, One:**● Anemia and, Both:****● Finding, ≥ One:**

- Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) ⁽²⁾

- Exertional dyspnea ⁽³⁾
- **Orthostatic hypotension, Both:**
 - Sustained drop in blood pressure within 3 min of sitting or standing
 - Systolic blood pressure drop ≥ 20 mmHg or diastolic blood pressure drop ≥ 10 mmHg
- Syncope ⁽⁴⁾
- **Hct or Hb monitoring at least 2x/24h and, Both:**
 - Blood product transfusion ⁽²⁾
 - Colonoscopy or specialist evaluation ≤ 24 h ⁽¹⁸⁾
- **ACUTE, One:** ⁽¹⁹⁾
 - Post Critical care ≤ 24 h
- **INTERMEDIATE, $\geq$ One:** ⁽²⁰⁾
 - NIPPV ⁽¹⁴⁾
 - Oxygen 35-39%(0.35-0.39) ⁽¹⁵⁾
- **CRITICAL, $\geq$ One:**
 - **Hemodynamic instability, Both:**
 - **Finding, $\geq$ Two:**
 - **Heart rate, One:** ⁽⁶⁾
 - > 180 /min, sustained (age ≤ 12 mos) ⁽⁷⁾
 - > 160 /min, sustained (age > 12 mos to 4 yrs) ⁽⁷⁾
 - > 150 /min, sustained (age > 4 yrs to 8 yrs) ⁽⁷⁾
 - > 140 /min, sustained (age > 8 yrs to 15 yrs) ⁽⁷⁾
 - > 130 /min, sustained (age > 15 yrs to < 18 yrs) ⁽⁷⁾
 - **Systolic BP, One:**
 - < 70 mmHg (age ≤ 12 mos)
 - < 80 mmHg (age > 1 to < 5 yrs)
 - < 90 mmHg (age 5 to < 18 yrs)
 - Labile ⁽⁸⁾
 - **MAP, One:** ⁽⁹⁾
 - < 55 mmHg (≤ 12 mos)
 - < 65 mmHg (> 1 to < 18 yrs)
 - **Intervention, $\geq$ One:**
 - Antifibrinolytic ⁽¹⁰⁾
 - Blood product transfusion ⁽¹¹⁾
 - IV fluid resuscitation ≤ 2 d ⁽¹²⁾
 - Hemodynamic monitoring, invasive ⁽¹³⁾
 - Mechanical ventilation
 - **NIPPV and, $\geq$ One:** ⁽¹⁴⁾
 - $\text{FiO}_2 \geq 40\%$ (0.40)
 - Respiratory intervention every 1-2h ⁽²¹⁾
 - Oxygen $\geq 40\%$ (0.40) ⁽¹⁵⁾
 - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
 - **Episode Day 3, One:**
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽¹⁶⁾
 - Hct or Hb within acceptable limits ⁽¹⁷⁾
 - Vital signs within acceptable limits ⁽¹⁷⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
 - For a condition other than anemia use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 24h, **All:** ⁽¹⁶⁾
 - Hct or Hb within acceptable limits ⁽¹⁷⁾

- Vital signs within acceptable limits ⁽¹⁷⁾
 - No complication or active comorbidity relevant to this episode of care ⁽²²⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **CRITICAL, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
-

Discharge, Level of Care, One: ⁽²³⁾

- **HOME, Both**:
 - Level of care appropriateness, **Both**:
 - Home environment safe and accessible ⁽²⁴⁾
 - Patient or caregiver demonstrates ability to manage care
 - Anemia discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽²⁵⁾
 - Education on the importance of, **All**:
 - Follow-up appointments
 - Knowing when to seek medical care
 - Obtaining flu and pneumococcal vaccinations
 - Medication reconciliation ⁽²⁶⁾
 - Follow-up care planned ⁽²⁷⁾
 - Identify and address transportation needs
- **HOME CARE, Both**:
 - Level of care appropriateness, **All**:
 - Home environment safe and accessible ⁽²⁴⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(28, 29)
 - Skilled services, **≥ One**: ⁽³⁰⁾
 - Skilled nursing ⁽³¹⁾
 - Skilled therapy (PT, OT, or ST)
 - Behavioral health ⁽³²⁾
 - Anemia discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽²⁵⁾
 - Education on the importance of, **All**:
 - Follow-up appointments
 - Knowing when to seek medical care
 - Obtaining flu and pneumococcal vaccinations
 - Medication reconciliation ⁽²⁶⁾
 - Follow-up care planned ⁽²⁷⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both**:
 - Level of care appropriateness, **One**:
 - Support systems available ⁽³³⁾
 - Skilled treatment, **≥ One**:

- Clinical assessment ⁽³⁴⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation ⁽³⁵⁾

● **SKILLED MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽³¹⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(36, 37)
- Functional impairments requiring at least supervision ⁽³⁸⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽²⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽³¹⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(36, 37)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁸⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽²⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
- Treatment precluded in a lower level

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(36, 37)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁸⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽²⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(36, 37)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁸⁾
- Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽³⁹⁾

- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽²⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁴⁰⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(41, 42)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁴³⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽²⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Notes:**1:**

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

Blood transfusion practice guidelines are intended to improve clinical outcomes while avoiding unnecessary transfusions that may not be clinically indicated. Recent high-quality evidence has resulted in guideline updates recommending lower hemoglobin transfusion thresholds for most hospitalized adult and pediatric patients. While individual institutional practice patterns may vary, the American Association of Blood Banks (AABB) recommendation for hemodynamically stable patients, including critically ill patients, is a restrictive transfusion threshold for a hemoglobin of less than 7 g/dL for most patients. A hemoglobin threshold of 7.5 g/dL is recommended for transfusion if patients are undergoing cardiac surgery and a threshold of 8 g/dL if patients are undergoing orthopedic surgery or have pre-existing cardiac disease. There are exceptions to these parameters including patients with acute coronary syndrome, severe thrombocytopenia, and chronic transfusion-dependent anemia (Carson et al., *Jama* 2023: E1-E11; Chegondi et al., *Transfus Med Hemother* 2016, 43: 297-301).

3:

Patients with severe anemia may exhibit exertional dyspnea (e.g., shortness of breath while walking up a flight of stairs) due to lack of circulating oxygen-carrying red blood cells to muscles. This may require hospitalization for blood transfusion depending on thresholds and possible need for supplemental oxygen (Yui and Brodsky, *Hematol Oncol Clin North Am* 2022, 36: 325-39; Long and Koyfman, *Emerg Med Clin North Am* 2018, 36: 609-30; DeVos and Jacobson, *Emerg Med Clin North Am* 2016, 34: 129-49).

4:

Syncope is the transient loss of consciousness caused by diminished cerebral blood flow, identified as brief, with spontaneous onset and recovery (Brignole et al., *Eur Heart J* 2018, 39: 1883-948; Shen et al., *Journal of the American College of Cardiology* 2017, 70: e39-e110).

5:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

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6:

In the pediatric patient, hemodynamic instability could be assessed by an elevated heart rate (HR) above the 95th percentile norms based on specific age groups. A combination of different parameters and their trends could also be utilized to ascertain impending clinical deterioration more accurately in these patients to allow for closer monitoring in the critical care setting, including respiratory rate, mental status, and capillary refill time, amongst others. Other factors that could impact HR in the pediatric population include pain, anxiety, medications, and/or fever. Although no specific consensus has been reached on HR thresholds that signify unstable hemodynamics by various pediatric societies, systemic reviews and observational studies have compared HR data to textbook references such as the Pediatric Advance Life Support (PALS). Eytan et al., in 2017, studied HR in pediatric patients who were critically ill in the pediatric intensive care unit (PICU) and developed percentiles of heart rates categorized into the same age groups used in previous studies, including PALS, with the goal of comparing their observed norms, especially between the 5th and 95th percentiles (Eytan et al., *Frontiers in Pediatrics* 2017, 5). Using their HR data above the 95th percentile, HR thresholds were utilized to identify the pediatric patient who may

require immediate or urgent intervention and rapidly treat unstable hemodynamics to prevent a potentially serious event.

7:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

8:

Blood pressure is considered labile when it is extremely variable and difficult to control.

9:

In pediatric patients, hemodynamic instability may be defined as a mean arterial pressure (MAP) less than 55 mmHg in infants up to 1 year of age and less than 65 mmHg in patients over 1 year of age (Matics and Sanchez-Pinto, JAMA Pediatr 2017, 171: e172352).

10:

Antifibrinolytic medications reduce blood loss by inhibiting clot degradation. Specific agents include tranexamic acid and aminocaproic acid, and may be indicated in bleeding due to traumatic, surgical, obstetric, gynecological, or gastrointestinal bleeding. Efficacy for these indications is in various stages of investigation, with the strongest evidence supporting use in bleeding due to trauma and surgery (Steinmetzer et al., J Med Chem 2020, 63: 1445-72; Draxler and Medcalf, Transfus Med Rev 2015, 29: 102-9; Hunt, Anaesthesia 2015, 70 Suppl 1: 50-3, e18).

11:

The volume of blood product infused in a pediatric patient is dependent on the weight of the child. For some children, small volume transfusions of blood products (especially packed red blood cells) are necessary to prevent fluid overload. An appropriate transfusion volume would be 10 to 15 mL/kg, infused at a rate of 2 to 4 mL/kg per hour (Park, In: Elsevier, ed. The Harriet Lane Handbook, 23 edn. 2024:365-407; Rodrigue and Sloan, In: Hematology, 8 edn. 2023:1892-9).

12:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine

output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

13:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

14:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

15:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air [room air: 21%(0.21) FIO₂] to correct hypoxia. For pediatric patients, the selection of an oxygen delivery system needs to take into consideration the patient's size, needs, and therapeutic goals. The oxygen concentration or percentage (FIO₂) delivered varies with the manufacturer's design, delivery device, patient's minute ventilation (dependent on the respiratory rate and tidal volume), and oxygen flow rate. In general, patients with higher minute ventilation (common in most sick patients and those with respiratory disease) require considerably higher flow rates to correct hypoxia. The following are examples of oxygen delivery systems commonly used for pediatric patients with the percentage range of FIO₂ optimally delivered:

Low-flow systems

- Nasal cannula / Nasopharyngeal catheters 24-35%(0.24-0.35) FIO₂

Reservoir systems (For neonates, infants, and young children, the maximum FIO₂ achievable is not well documented)

- Simple oxygen masks 35-50%(0.35-0.50) FIO₂
- Partial-rebreathing masks 40-60%(0.40-0.60) FIO₂
- Non-rebreathing masks > 60%(0.60) FIO₂

High-flow systems

- Air-entrainment masks / nebulizers 24-40%(0.24-0.40) FIO₂
- Heated, humidified, high flow oxygen concentrating nasal cannula 21-100%(0.21-1.0) FIO₂

Enclosure systems

- Oxygen hoods / tents 21-100%(0.21-1.0) FIO₂

For children less than 5 years old, the oxygen regimen may need to be adjusted to a lesser concentration and flow rate, while still requiring acute level of care due to the patient's size, metabolic needs, and vital sign instability.

16:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for

transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

17:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

18:

Specialist evaluation refers to an initial consultation by a medical professional certified in a specialty area (e.g., pediatric endocrine, cardiac or neurology). A pediatric hematology consult may be warranted when patients are not responding to standard treatment in the expected time frame (Verbillion and Dupre, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:1462-79).

19:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

20:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

21:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

22:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

23:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

24:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

25:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

26:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

27:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

28:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

29:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

30:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

31:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

32:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

33:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

34:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

35:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

36:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

37:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

38:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs

- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

39:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

40:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

41:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

42:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

43:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.